

Business Analytics Case Study

Understanding Usage Patterns in a Bike-Share Service

Vanessa Negreiros

Business Analytics | Data Analysis | Tableau

December 2025

1. Executive Overview

This case study analyzes usage patterns of a bike-share service to understand how annual members and casual riders differ in behavior, preferences, and usage context.

The objective is to support data-driven business decisions, particularly strategies aimed at increasing membership adoption and long-term customer value.

A structured analytics approach was applied to transform raw operational data into clear insights and actionable recommendations.

2. Business Context & Objective

Bike-share services operate at the intersection of urban mobility, convenience, and sustainability.

Understanding how different user segments interact with the service is critical for designing effective pricing models, marketing strategies, and service improvements.

Business question: How do annual members and casual riders use the service differently, and how can these insights support membership growth strategies?

3. Data & Analytical Approach

The analysis is based on historical trip data from two distinct periods, covering more than 770,000 rides.

The process included data validation, cleaning, harmonization of datasets, and the creation of derived metrics to support business-oriented analysis.

4. Key Insights

User Distribution: Annual members account for approximately 93% of total trips, while casual riders represent a smaller share of overall usage.

Trip Duration: Members demonstrate more consistent usage patterns, while casual riders show more sporadic behavior with wider variation in trip length.

Weekly Trends: Members ride consistently on weekdays, indicating commuting behavior, whereas casual riders concentrate trips on weekends.

Station Preferences: Casual riders tend to start trips near tourist and recreational areas, while members favor stations in business districts.

5. Business Implications

The findings confirm that members and casual riders engage with the service in systematically different ways.

Casual riders represent a clear opportunity for growth if the service is positioned as a practical daily mobility solution rather than purely recreational.

6. Strategic Recommendations

Position memberships as a convenient commuting alternative, emphasizing time savings, sustainability, and health benefits.

Use targeted digital and location-based campaigns in high-traffic recreational areas to promote membership plans.

Leverage station-level insights to support conversion-focused marketing initiatives.

7. Deliverables

This executive case study document.

An interactive Tableau dashboard presenting key metrics and visual insights.

8. Final Note

This project demonstrates the application of business-oriented analytics, combining data preparation, analysis, visualization, and strategic thinking to support informed decision-making.